

PRODUCT DATA INSIGHTS

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DATE : 31/MAY/2026

Project Objectives

The objective of this project is to analyze product data using Microsoft Excel to demonstrate data cleaning, validation, and visualization techniques. The dataset includes details such as Product ID, Manufacturing Date, Country Code, Product Name, Brand Name, Price, Quantity, Category, and Total Price. The goal is to apply Excel formulas and logical functions to handle missing values, correct category mismatches, and compute total prices.

Dataset Description

The dataset contains detailed information about various products sold across multiple countries. Each record represents a unique product identified by its Product ID, which includes the date and country code. The dataset captures essential attributes such as Manufacturing Date, Product Name, Brand Name, Price (\$), Quantity, Category, and Total Price.

Column Description

Column Name	Description
Product ID	A unique identifier combining the manufacturing date and country code (e.g., 28-JAN-US).
Manufacturing Date	The date when the product was manufactured. Used to analyze production timelines.
Country Code	Represents the country where the product was produced or sold (e.g., US, UK, IN, AU).
Product Name	The name of the product (e.g., Laptop, Sneakers, Coffee Maker).
Brand Name	The manufacturer or brand associated with the product (e.g., Dell, Nike, Keurig).
Product Brand	A combined field showing both product and brand (e.g., Laptop–Dell). Useful for validation checks.

Price (\$)	The unit price of the product in dollars. Used for cost and pricing analysis.
Quantity	The number of units sold or available. Helps calculate total revenue and stock levels.
Category	The product type or segment (e.g., Electronics, Fashion, Kitchen). Used for classification and trend analysis.
Total Price	Computed as Price × Quantity. Represents total sales value per product entry.

Tasks:

1. Handling Missing Values :

- a) Check for missing values in the “Price Column”.

Formula=IF(ISBLANK(G2),AVERAGE(G2:G32),G2)

Result= ₹ 1,000

- b) Products with missing Categories.

Formula=IF(ISBLANK(I2),"Unknown",I2)

Result= Electronics

2. Correcting Inconsistent Data :

- a) Identify any inconsistent text format present in the “Product Name” column.

Formula=PROPER(TRIM(CLEAN(D2)))

Result= Laptop

- b) Identify any typos present in the “Category” column.

First=TRIM(CLEAN(I2))

Electronics

Second=Find and Replace

Find what= Electroni - Replace with =Electronics

- c) Find and Replace function to standardize the text format in the “Product Name”

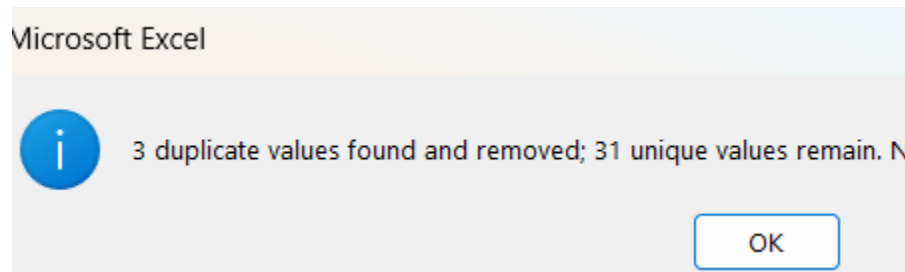
Product Name column -Find and Replace

laptop=Laptop

Fix any typos in “Category “ column

Find what= Electroni -Replace with =Electronics

3. Removing Duplicates :



4. Splitting and Merging Data :

a) Split the "Product ID" column into two separate columns for " Manufacturing Date" and "Country Code". Remove unnecessary characters, if any.

B	C
Manufacturing Date	Contry Code
28-Jan-00	US
15-Jan-00	US
3-Jan-00	US
11-Jan-00	US
22-Jan-00	US
7-Jan-00	UK
19-Jan-00	UK
23-Jan-00	UK
5-Jan-00	UK

Year is not mentioned in Dataset.so I considered as 28-jan-00.

b) Merge the "Brand Name" and "Product Name" columns into one column named "Product Brand".

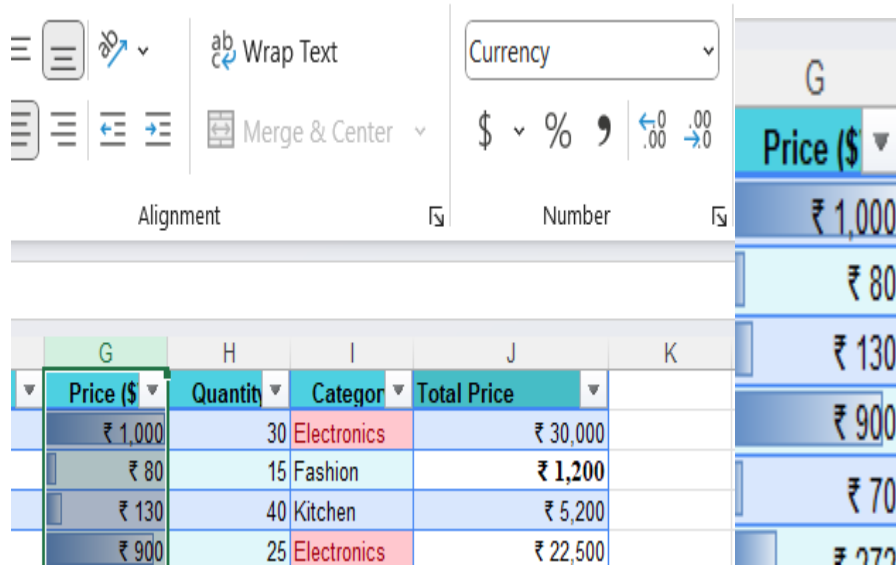
Formula =CONCATENATE(D2,"--",E2)

Result =Laptop--Dell

F
Product Brand
Laptop--Dell
Sneakers--Nike
Coffee Maker--Keurig
Smartphone--Samsung
Backpack--North Face
Headphones--Sony

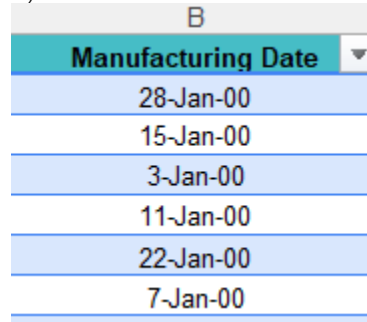
5. Number Formatting :

- a) Format the data type of the "Price" column to a currency format.



	G	H	I	J	K
	Price (\$)	Quantity	Category	Total Price	
	₹ 1,000	30	Electronics	₹ 30,000	
	₹ 80	15	Fashion	₹ 1,200	
	₹ 130	40	Kitchen	₹ 5,200	
	₹ 900	25	Electronics	₹ 22,500	

- b) Format the "Manufacturing Date" column to display dates in the "DD-MM-YYYY " format.

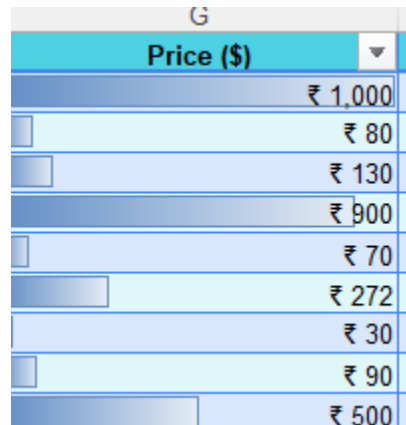


B
Manufacturing Date
28-Jan-00
15-Jan-00
3-Jan-00
11-Jan-00
22-Jan-00
7-Jan-00

Year is not mentioned in Dataset.

6. Conditional Formatting

- a) Applied data bar or color scales conditional formatting in the "Price" column.



G
Price (\$)
₹ 1,000
₹ 80
₹ 130
₹ 900
₹ 70
₹ 272
₹ 30
₹ 90
₹ 500

b) Create a custom rule for conditional formatting in the "Category" column to highlight cells where the category is "Electronics."

Category
Electronics
Fashion
Kitchen
Electronics
Unknown
Electronics
Fashion

Home—conditional formatting--highlight rules.

Conclusion

This Excel-based project successfully demonstrates how data cleaning, validation, and analysis can transform raw product information into meaningful insights. Through systematic correction of inconsistent text formats, handling of missing values, and category validation, the dataset was refined for accurate computation of total prices and improved readability.

Thank You
Entri Elevate